



Internship

Day 6

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DGL05

nVIDIA

Faculties

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Day 6 — What Transformers Do

THEORY (30 min): What transformers do (intuition)

HANDS-ON: Text/sentiment classification with a pretrained Hugging Face model → use transformer embeddings for a custom classifier → evaluate with precision/recall/F1.

DELIVERABLE: A working, evaluated NLP app

What Are Large Language Models (LLMs)?

- ❖ Large Language Models (LLMs) are a type of AI model that has been trained on vast amounts of text data.
- ❖ These models are designed to **understand, generate, and predict text.**
- ❖ By analyzing patterns in the data, **LLMs can generate coherent and contextually appropriate responses to text-based inputs.**

What Are Transformers?

- Transformers are a type of neural network architecture that has revolutionized the field of natural language processing (NLP).
- Introduced in a 2017 **paper titled "Attention is All You Need"** by Vaswani et al.,
- Transformers are designed to handle sequential data, such as text, by using a mechanism called self-attention.

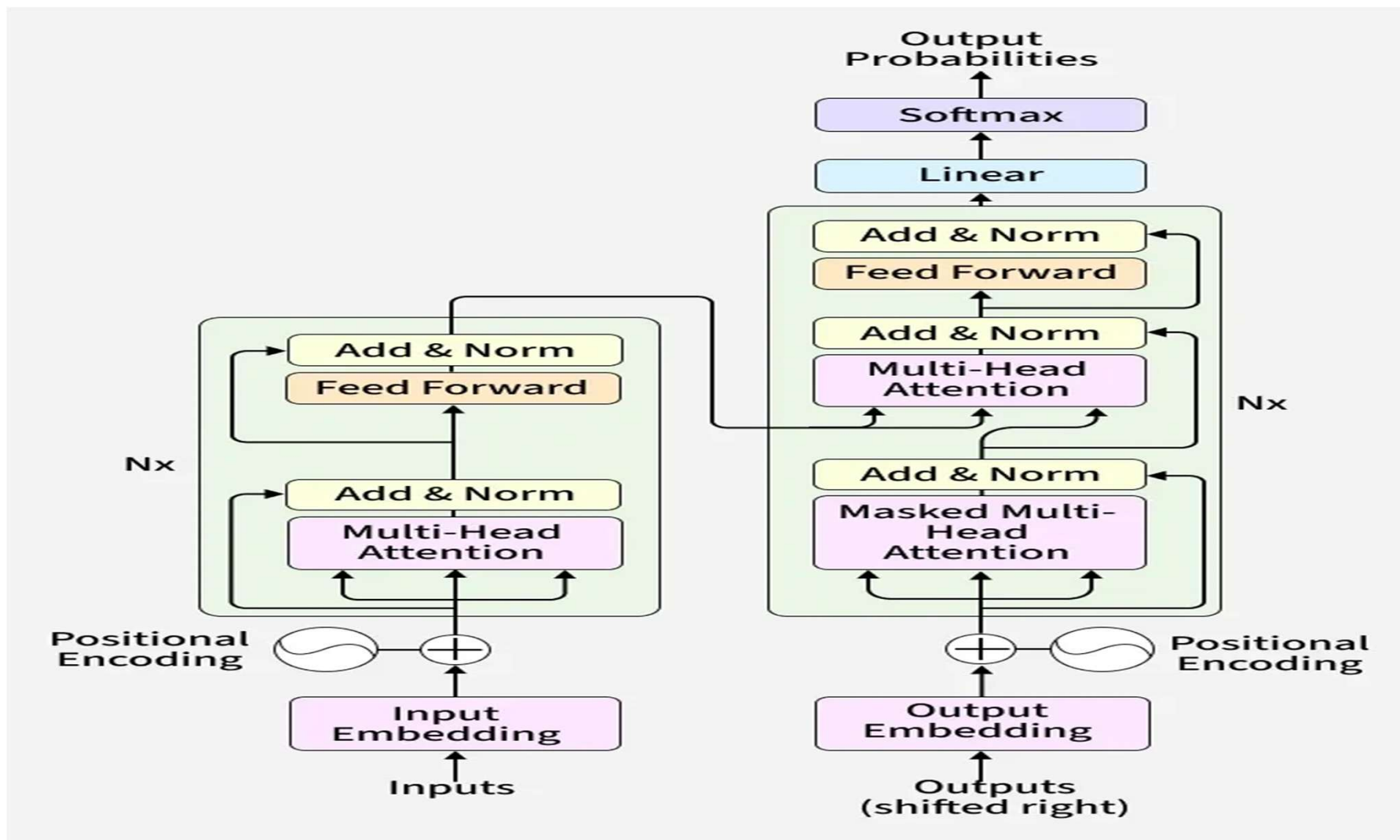
Key Characteristics of Transformers:

- ❖ **Self-Attention Mechanism:** The [self-attention mechanism](#) allows Transformers to **weigh the importance of different words in a sentence when making predictions**. This is crucial for understanding context, especially in long sentences.
- ❖ **Parallelization:** Unlike traditional RNNs (Recurrent Neural Networks), which process data sequentially, Transformers can process multiple words at once, making them faster and more efficient.
- ❖ **Versatility:** **Transformers are not limited to language tasks; they can be applied to any problem involving sequential data, including tasks like image recognition and time-series forecasting.**

Applications

- ❑ **LLMs:** Used for a wide range of NLP tasks, **from text generation and summarization to translation and sentiment analysis.**
- ❑ **Transformers:** Employed not just in NLP but also in other **areas requiring the processing of sequential data, such as speech recognition and computer vision.**

Aspect	Large Language Models (LLMs)	Transformers
Purpose	Generate and understand natural language	Neural network architecture for sequential data
Architecture	Often based on the Transformer architecture	Uses self-attention and parallel processing
Applications	NLP tasks like text generation, summarization, etc.	NLP, image recognition, speech recognition, etc.
Training	Requires massive datasets and resources, often using Transformers	Adaptable to various tasks with different types of sequential data
Output	Human-like text	Task-dependent output (text, predictions, etc.)



Input Processing

Tokenization & Embeddings: Input **text is broken down into tokens (words or subwords)** and mapped into numerical vectors that capture their semantic meaning.

Positional Encoding: Because transformers process all words simultaneously and **lack inherent sequence awareness, they add a positional encoding to the embeddings** so the model understands the order of the words.

Encoder

The encoder reads the input sequence and generates a rich, context-aware representation. It primarily contains:

Self-Attention: Each token "looks" at all other tokens in the sequence to gather context (e.g., in "The bank of the river," the model understands "bank" refers to land, not money).

Multi-Head Attention: Runs multiple attention mechanisms in parallel, allowing the model to focus on different aspects of a word's meaning simultaneously (e.g., grammatical vs. semantic relationships).

Feed-Forward Networks & Residuals: The attention outputs go through a position-wise, fully-connected neural network to refine the token representations, accompanied by residual connections and layer normalization for stable training.

Decoder

The decoder takes the output of the encoder and iteratively generates the target sequence, such as translating text or continuing a story. It includes:

Masked Self-Attention: Similar to the encoder but blocks the model from "**looking ahead**" at future tokens, ensuring predictions are based solely on past contexts.

Encoder-Decoder Attention: Allows the decoder to focus on the **most relevant parts of the input sequence processed by the encoder.**

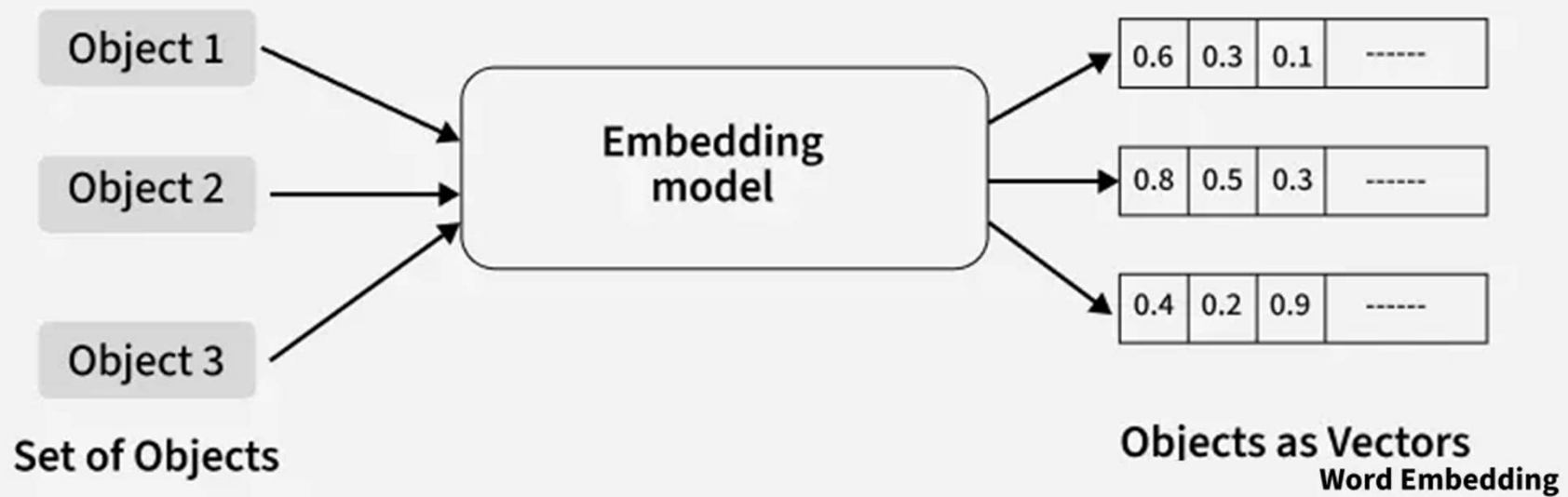
Word Embeddings in NLP

- **Word Embeddings are numeric representations of words** in a lower-dimensional space, that capture semantic and syntactic information.
- They play an important role in Natural Language Processing (NLP) tasks.
- Some traditional and neural approaches used to implement **Word Embeddings, such as TF-IDF, Word2Vec, and GloVe.**

How it Helps?

- ❖ To **reduce dimensionality**.
- ❖ To use a word to **predict the words around** it.
- ❖ Helps in enhancing model **interpretability** due to numerical representation.
- ❖ **Inter-word semantics and similarity** can be captured.

Embeddings in Natural Language Processing



Cat →	1.2	-0.1	4.3	3.2
Mat →	0.4	2.5	-0.9	0.5
on →	2.1	0.3	0.1	0.4

Happy			
Sad			
Excited			
Sick			

The emoji vectors for the emojis will be:
[happy, sad, excited, sick]

 = [1,0,1,0]

 = [0,1,0,1]

 = [0,1,0,1]

...

Techniques in Embeddings in NLP

Types Of Word Embeddings

Frequency Based Embeddings

Bag Of Words

TF-IDF

Co-occurrence Matrices

Prediction Based Word Embedding

Word2Vec (CBOW, Skip-gram)

GloVe

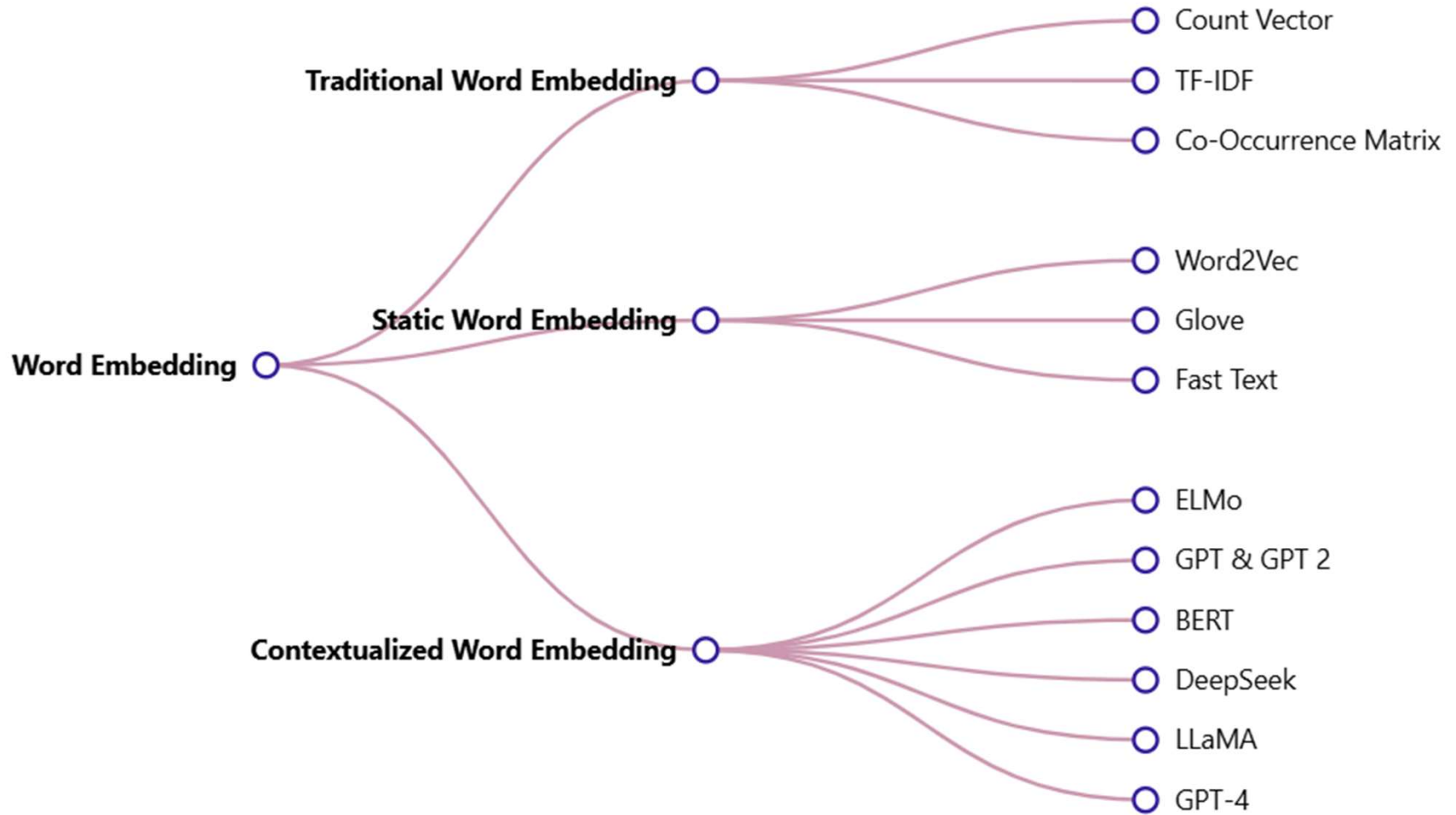
FastText

Contextualized Word Embeddings

ELMO

BERT

GPT



Types

Today, the architecture is typically categorized into three distinct variations depending on the use case:

1. **Encoder-only models** (like BERT) for **classification and text analysis**.
2. **Decoder-only models** (like GPT) optimized for **generative AI and predictive text**.
3. **Encoder-decoder models** (like original transformer/T5) used for translation or text-to-text tasks.

Examples of Transformers

1. BERT (Bidirectional Encoder Representations from Transformers):

Use Case: Text classification, sentiment analysis, question answering, named entity recognition.

Characteristic: Focuses on understanding the context of a word in a sentence by looking at both its left and right context.

2. GPT (Generative Pre-trained Transformer):

Use Case: Text generation, summarization, translation.

Characteristic: Uses the Transformer architecture's decoder to generate coherent and contextually relevant text.

3. T5 (Text-To-Text Transfer Transformer):

Use Case: Translation, text summarization, text generation.

Characteristic: Converts every NLP problem into a text-to-text format, enabling a single model to handle multiple tasks.

4. ViT (Vision Transformer):

Use Case: Image classification, object detection.

Characteristic: Adapts the Transformer architecture for processing images, treating image patches as sequence data.

5. RoBERTa (Robustly Optimized BERT Pretraining Approach):

Use Case: Similar to BERT, with improvements in training methodology.

Characteristic: Focuses on better performance by optimizing the pre-training process.

THANK YOU